第 14 章　二重积分

1.【答案】(B)

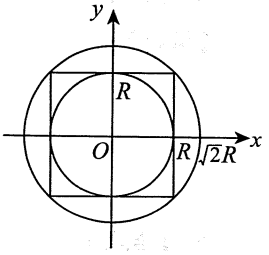
【解析】当 $\frac{1}{2}\leqslant t\leqslant 1$ 时，$\ln t<\sin t<t\leqslant 1$，故当 $\frac{1}{2}\leqslant x+y\leqslant 1$ 时，有

$$\ln^7(x+y)<\sin^7(x+y)<(x+y)^7,$$

根据二重积分比较定理，可知 $I_1<I_3<I_2$，故选(B).

2.【答案】(C)

【解析】D_1，D_2 是以原点为圆心、半径分别为 R 与 $\sqrt{2}R$ 的圆，D_3 是正方形，边长为 $2R$，如图所示.显然有



$$D_1\subset D_3\subset D_2.$$

由于 $e^{-(x^2+y^2)}$ 恒正，因此(C)成立.

3.【答案】(B)

【解析】由积分中值定理知，存在 $(\xi,\eta)\in\left\{(x,y)\,\middle|\,x^2+y^2\leqslant r^2\right\}$，使得

$$\lim_{r\to 0}\frac{1}{\pi r^2}\iint\limits_{x^2+y^2\leqslant r^2}e^{x^2-y^2}\cos(x+y)\mathrm{d}x\mathrm{d}y=\lim_{r\to 0}\frac{1}{\pi r^2}\cdot e^{\xi^2-\eta^2}\cos(\xi+\eta)\cdot\pi r^2$$

$$=\lim_{r\to 0}e^{\xi^2-\eta^2}\cos(\xi+\eta)=\lim_{(\xi,\eta)\to(0,0)}e^{\xi^2-\eta^2}\cos(\xi+\eta)=1.$$

4.【答案】(C)

【解析】
$$\iint\limits_D(axy+by^2)\mathrm{d}x\mathrm{d}y=\iint\limits_D axy\mathrm{d}x\mathrm{d}y+\iint\limits_D by^2\mathrm{d}x\mathrm{d}y=I_1+I_2,$$

注意到积分区域 D 关于 y 轴对称，函数 axy 关于 x 为奇函数，所以 $I_1=0$. 因此原积分的符号只与 b 有关，而与 a 无关.

5.【答案】$e^{-\frac{1}{2}}$

【解析】交换积分次序，得

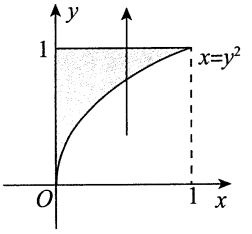
$$\text{原式}=\int_0^1\mathrm{d}y\int_{y^2}^1 e^{-\frac{y^2}{2}}\mathrm{d}x=\int_0^1(1-y^2)e^{-\frac{y^2}{2}}\mathrm{d}y=\int_0^1 e^{-\frac{y^2}{2}}\mathrm{d}y+\int_0^1 y\mathrm{d}\left(e^{-\frac{y^2}{2}}\right)$$

$$=\int_0^1 e^{-\frac{y^2}{2}}\mathrm{d}y+ye^{-\frac{y^2}{2}}\Big|_0^1-\int_0^1 e^{-\frac{y^2}{2}}\mathrm{d}y=e^{-\frac{1}{2}}.$$

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6.【答案】1

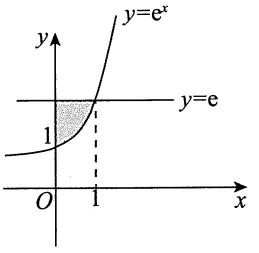
【解析】积分区域如图所示，则



$$\begin{aligned}\text{原式}&=\int_0^1 \mathrm{d}x\int_{\sqrt{x}}^1 \frac{x\mathrm{e}^x}{1-\sqrt{x}}\mathrm{d}y\\&=\int_0^1 x\mathrm{e}^x\mathrm{d}x=(x-1)\mathrm{e}^x\Big|_0^1\\&=1.\end{aligned}$$

7.【答案】(B)

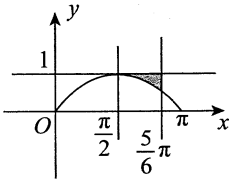
【解析】积分区域如图所示，所以先积 y 时应是从 $y=\mathrm{e}^x$ 到 $y=\mathrm{e}$，故



$I=\int_0^1 \mathrm{d}x\int_{\mathrm{e}^x}^{\mathrm{e}} f(x,y)\mathrm{d}y$，选(B).

8.【答案】(B)

【解析】积分区域如图所示，交换积分次序，得

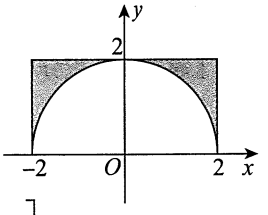


$$\int_{\frac{\pi}{2}}^{\frac{5\pi}{6}}\mathrm{d}x\int_{\sin x}^1 f(x,y)\mathrm{d}y=\int_{\frac{1}{2}}^1\mathrm{d}y\int_{\pi-\arcsin y}^{\frac{5\pi}{6}} f(x,y)\mathrm{d}x,$$

故选(B).

9.【答案】(C)

【解析】由题意知，此二重积分的积分区域为



$$D=\left\{(x,y)\middle|\sqrt{4-x^2}\leqslant y\leqslant 2,-2\leqslant x\leqslant 2\right\},$$

对应图像如图所示.

$$I=\int_{-2}^2\mathrm{d}x\int_{\sqrt{4-x^2}}^2 f(x,y)\mathrm{d}y=\int_0^2\left[\int_{-2}^{-\sqrt{4-y^2}} f(x,y)\mathrm{d}x+\int_{\sqrt{4-y^2}}^2 f(x,y)\mathrm{d}x\right]\mathrm{d}y.$$

故选项(A)错误.

又 $f(x,y)=f(-x,y)$，且积分区域关于 y 轴对称，则 $I=2\int_0^2\left[\int_{\sqrt{4-y^2}}^2 f(x,y)\mathrm{d}x\right]\mathrm{d}y$. 故选项(B)错误.

记 $D_1=\left\{(x,y)\middle|(x,y)\in D,x>y\right\}$. 由 $f(x,y)=f(y,x)$，得 $I=4\iint\limits_{D_1} f(x,y)\mathrm{d}x\mathrm{d}y$.

用极坐标来表示，$I=4\int_0^{\frac{\pi}{4}}\mathrm{d}\theta\int_2^{2\sec\theta} f(r\cos\theta,r\sin\theta)r\mathrm{d}r$. 故选(C).

10.【答案】$\dfrac{4}{27}$

【解析】
$$\iint\limits_D\sqrt{x^2+y^2}\mathrm{d}x\mathrm{d}y=\int_0^{\frac{\pi}{3}}\mathrm{d}\theta\int_0^{\sin 3\theta}\sqrt{r^2\cos^2\theta+r^2\sin^2\theta}\cdot r\mathrm{d}r$$

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$$=\int_0^{\frac{\pi}{3}}\mathrm{d}\theta\int_0^{\sin 3\theta}r^2\mathrm{d}r=\frac{1}{3}\int_0^{\frac{\pi}{3}}\sin^3 3\theta\mathrm{d}\theta$$

$$=\frac{1}{9}\int_0^{\pi}\sin^3 t\mathrm{d}t=\frac{2}{9}\int_0^{\frac{\pi}{2}}\sin^3 t\mathrm{d}t=\frac{2}{9}\cdot\frac{2}{3}=\frac{4}{27}.$$

11.【答案】 $\dfrac{5}{144}$

【解析】由被积函数知，应先对 x 积分，再对 y 积分.

$$\iint_D xe^{-y^2}\mathrm{d}x\mathrm{d}y=\int_0^{+\infty}e^{-y^2}\mathrm{d}y\int_{\frac{\sqrt{y}}{3}}^{\frac{\sqrt{y}}{2}}x\mathrm{d}x=\int_0^{+\infty}\frac{1}{2}\left(\frac{y}{4}-\frac{y}{9}\right)e^{-y^2}\mathrm{d}y=\frac{5}{144}(-e^{-y^2})\Big|_0^{+\infty}=\frac{5}{144}.$$

12.【答案】 $\dfrac{4}{9}a^3(3\pi-4)$

【解析】令 $x=r\cos\theta$，$y=r\sin\theta$，则

$$\iint_D\sqrt{4a^2-x^2-y^2}\mathrm{d}x\mathrm{d}y=\int_0^{\frac{\pi}{2}}\mathrm{d}\theta\int_0^{2a\cos\theta}\sqrt{4a^2-r^2}r\mathrm{d}r$$

$$=\int_0^{\frac{\pi}{2}}\left[-\frac{1}{3}(4a^2-r^2)^{\frac{3}{2}}\Big|_0^{2a\cos\theta}\right]\mathrm{d}\theta$$

$$=\frac{8a^3}{3}\int_0^{\frac{\pi}{2}}(1-\sin^3\theta)\mathrm{d}\theta$$

$$=\frac{4}{9}a^3(3\pi-4).$$

13.【解析】令 $x=r\cos\theta$，$y=r\sin\theta$，由 $y=\sqrt{3(1-x^2)}$，得 $r^2\sin^2\theta+3r^2\cos^2\theta=3$，于是 $r^2=\dfrac{3}{3\cos^2\theta+\sin^2\theta}$，积分区域如图所示，则

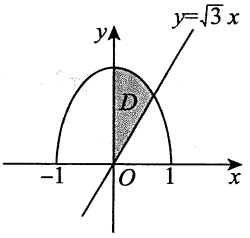
$$\iint_D(x^2+y^2)\mathrm{d}x\mathrm{d}y=\int_{\frac{\pi}{3}}^{\frac{\pi}{2}}\mathrm{d}\theta\int_0^{\sqrt{\frac{3}{3\cos^2\theta+\sin^2\theta}}}r^3\mathrm{d}r=\frac{9}{4}\int_{\frac{\pi}{3}}^{\frac{\pi}{2}}\frac{\tan^2\theta+1}{(3+\tan^2\theta)^2}\mathrm{d}(\tan\theta)$$

$$=\frac{9}{4}\int_{\sqrt{3}}^{+\infty}\frac{t^2+1}{(3+t^2)^2}\mathrm{d}t=\frac{9}{4}\int_{\sqrt{3}}^{+\infty}\frac{t^2+3-2}{(t^2+3)^2}\mathrm{d}t$$

$$=\frac{9}{4}\int_{\sqrt{3}}^{+\infty}\frac{\mathrm{d}t}{t^2+3}-\frac{9}{4}\int_{\sqrt{3}}^{+\infty}\frac{2\mathrm{d}t}{(t^2+3)^2}$$

$$=\frac{9}{4}\cdot\frac{1}{\sqrt{3}}\arctan\frac{t}{\sqrt{3}}\Big|_{\sqrt{3}}^{+\infty}-\frac{1}{4}\left(\frac{3t}{t^2+3}+\sqrt{3}\arctan\frac{t}{\sqrt{3}}\right)\Big|_{\sqrt{3}}^{+\infty}$$

$$=\frac{3\sqrt{3}\pi}{16}-\frac{(\pi-2)\sqrt{3}}{16}=\frac{\sqrt{3}}{8}(1+\pi).$$



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14.【解析】由轮换对称性，得

$$\iint_D \frac{x\cos\sqrt{x^2+y^2}}{x+y}\,\mathrm{d}x\mathrm{d}y=\iint_D \frac{y\cos\sqrt{x^2+y^2}}{x+y}\,\mathrm{d}x\mathrm{d}y$$

$$=\frac{1}{2}\iint_D \cos\sqrt{x^2+y^2}\,\mathrm{d}x\mathrm{d}y=\frac{1}{2}\int_0^{\frac{\pi}{2}}\mathrm{d}\theta\int_1^2 r\cos r\,\mathrm{d}r$$

$$=\frac{\pi}{4}(2\sin 2+\cos 2-\sin 1-\cos 1).$$

15.【解析】由对称性，得

$$\iint_D \frac{x^2-x\cos y-y^2}{x^2+y^2}\,\mathrm{d}x\mathrm{d}y=\iint_D \frac{x^2-y^2}{x^2+y^2}\,\mathrm{d}x\mathrm{d}y=2\int_{\frac{\pi}{4}}^{\frac{\pi}{2}}\mathrm{d}\theta\int_0^{\frac{1}{\sin\theta}}\frac{r^2\cos^2\theta-r^2\sin^2\theta}{r^2}\,r\mathrm{d}r$$

$$=\int_{\frac{\pi}{4}}^{\frac{\pi}{2}}(\cos^2\theta-\sin^2\theta)\frac{1}{\sin^2\theta}\,\mathrm{d}\theta=\int_{\frac{\pi}{4}}^{\frac{\pi}{2}}\cot^2\theta\mathrm{d}\theta-\int_{\frac{\pi}{4}}^{\frac{\pi}{2}}1\mathrm{d}\theta$$

$$=\int_{\frac{\pi}{4}}^{\frac{\pi}{2}}\csc^2\theta\mathrm{d}\theta-\int_{\frac{\pi}{4}}^{\frac{\pi}{2}}2\mathrm{d}\theta=-\cot\theta\Big|_{\frac{\pi}{4}}^{\frac{\pi}{2}}-\frac{\pi}{2}=1-\frac{\pi}{2}.$$

16.【解析】积分区域如图所示，由对称性，用极坐标，得

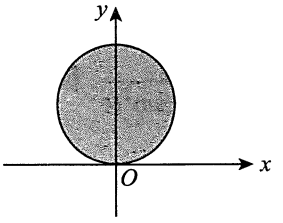
$$\iint_D \sqrt{1-x^2-y^2}\,\mathrm{d}\sigma=2\int_0^{\frac{\pi}{2}}\mathrm{d}\theta\int_0^{\sin\theta}\sqrt{1-r^2}\,r\mathrm{d}r$$

$$=-\int_0^{\frac{\pi}{2}}\mathrm{d}\theta\int_0^{\sin\theta}(1-r^2)^{\frac{1}{2}}\mathrm{d}(1-r^2)$$

$$=-\int_0^{\frac{\pi}{2}}\left[\frac{2}{3}(1-r^2)^{\frac{3}{2}}\Big|_0^{\sin\theta}\right]\mathrm{d}\theta$$

$$=-\frac{2}{3}\int_0^{\frac{\pi}{2}}(\cos^3\theta-1)\mathrm{d}\theta$$

$$=-\frac{2}{3}\left(\frac{2}{3}-\frac{\pi}{2}\right)=\frac{\pi}{3}-\frac{4}{9}.$$



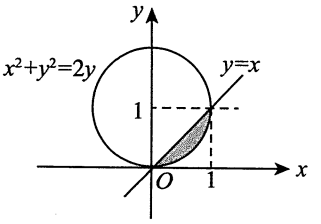
17.【解析】由于$f(x)+f(-x)$为偶函数，因而$x[f(x)+f(-x)]$为奇函数，又$f(x)-f(-x)$为奇函数，从而$(x+1)f(x)+(x-1)f(-x)=x[f(x)+f(-x)]+[f(x)-f(-x)]$为奇函数.故

$$\text{原式}=\int_{-1}^{1}[(x+1)f(x)+(x-1)f(-x)]\,\mathrm{d}x\int_{x^3}^{1}\sin y\,\mathrm{d}y$$

$$=\int_{-1}^{1}(\cos x^3-\cos 1)[(x+1)f(x)+(x-1)f(-x)]\,\mathrm{d}x=0.$$

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18.【解析】积分区域如图所示，则

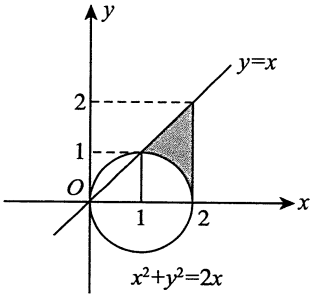


$$I=\int_0^1 \mathrm{d}y\int_y^{\sqrt{2y-y^2}} \frac{1}{\sqrt{x^2+y^2}\sqrt{4-x^2-y^2}}\,\mathrm{d}x$$

$$=\int_0^{\frac{\pi}{4}}\mathrm{d}\theta\int_0^{2\sin\theta}\frac{1}{\sqrt{4-r^2}}\,\mathrm{d}r=\int_0^{\frac{\pi}{4}}\left(\arcsin\frac{r}{2}\Big|_0^{2\sin\theta}\right)\mathrm{d}\theta$$

$$=\int_0^{\frac{\pi}{4}}\theta\,\mathrm{d}\theta=\frac{1}{2}\theta^2\Big|_0^{\frac{\pi}{4}}=\frac{\pi^2}{32}.$$

19.【解析】被积函数$f(x,y)$的非零区域和积分区域的公共部分如图所示，故



$$\iint_D f(x,y)\mathrm{d}\sigma=\int_0^{\frac{\pi}{4}}\mathrm{d}\theta\int_{2\cos\theta}^{\frac{2}{\cos\theta}} r\sin\theta\cdot r\,\mathrm{d}r$$

$$=\frac{1}{3}\int_0^{\frac{\pi}{4}}\left(\sin\theta\cdot r^3\Big|_{r=2\cos\theta}^{r=\frac{2}{\cos\theta}}\right)\mathrm{d}\theta$$

$$=\frac{8}{3}\int_0^{\frac{\pi}{4}}\left(\frac{\sin\theta}{\cos^3\theta}-\cos^3\theta\sin\theta\right)\mathrm{d}\theta$$

$$=\frac{8}{3}\left(\frac{1}{2}\frac{1}{\cos^2\theta}+\frac{1}{4}\cos^4\theta\right)\Big|_0^{\frac{\pi}{4}}$$

$$=\frac{5}{6}.$$

20.【解析】方法一　用极坐标系.

$$\text{原式}=\int_0^{\frac{\pi}{2}}\mathrm{d}\theta\int_0^{\frac{1}{\cos\theta+\sin\theta}}\frac{\mathrm{e}^{-r(\cos\theta+\sin\theta)}}{\sqrt{\cos\theta\sin\theta}}\,\mathrm{d}r$$

$$=\int_0^{\frac{\pi}{2}}\left[\frac{1}{\sqrt{\cos\theta\sin\theta}}\cdot\frac{-1}{\cos\theta+\sin\theta}\cdot\mathrm{e}^{-r(\cos\theta+\sin\theta)}\Big|_{r=0}^{r=\frac{1}{\cos\theta+\sin\theta}}\right]\mathrm{d}\theta$$

$$=\left(1-\frac{1}{\mathrm{e}}\right)\int_0^{\frac{\pi}{2}}\frac{1}{(\cos\theta+\sin\theta)\sqrt{\cos\theta\sin\theta}}\,\mathrm{d}\theta$$

$$=\left(1-\frac{1}{\mathrm{e}}\right)\int_0^{\frac{\pi}{2}}\frac{\mathrm{d}\theta}{\cos^2\theta(1+\tan\theta)\sqrt{\tan\theta}}$$

$$=\left(1-\frac{1}{\mathrm{e}}\right)\int_0^{\frac{\pi}{2}}\frac{\mathrm{d}(\tan\theta)}{(1+\tan\theta)\sqrt{\tan\theta}}$$

$$\xlongequal{\tan\theta=u}\left(1-\frac{1}{\mathrm{e}}\right)\int_0^{+\infty}\frac{\mathrm{d}u}{(1+u)\sqrt{u}}$$

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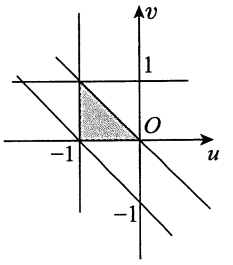
$$=\left(1-\frac{1}{\mathrm{e}}\right)\cdot 2\int_0^{+\infty}\frac{\mathrm{d}(\sqrt{u})}{1+(\sqrt{u})^2}$$

$$=2\left(1-\frac{1}{\mathrm{e}}\right)\cdot \arctan\sqrt{u}\Big|_0^{+\infty}$$

$$=\pi\left(1-\frac{1}{\mathrm{e}}\right).$$

方法二 用换元法.

令$\begin{cases}u=-(x+y),\\ v=x,\end{cases}$则$D'=\{(u,v)\,|\,-1\leqslant u\leqslant 0,0\leqslant v\leqslant 1,-1\leqslant u+v\leqslant 0\}$，区域$D'$如图中阴影部分所示.



$$\text{原式}=\iint_{D'}\frac{\mathrm{e}^u}{\sqrt{v(-v-u)}}\left\|\begin{matrix}\frac{\partial x}{\partial u} & \frac{\partial x}{\partial v}\\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v}\end{matrix}\right\|\mathrm{d}u\mathrm{d}v=\iint_{D'}\frac{\mathrm{e}^u}{\sqrt{v(-v-u)}}\mathrm{d}u\mathrm{d}v$$

$$=\int_{-1}^{0}\mathrm{d}u\int_0^{-u}\frac{\mathrm{e}^u}{\sqrt{-v^2-uv}}\mathrm{d}v=\int_{-1}^{0}\mathrm{e}^u\left(\arcsin\frac{v+\frac{u}{2}}{-\frac{u}{2}}\Big|_{v=0}^{v=-u}\right)\mathrm{d}u$$

$$=\int_{-1}^{0}\mathrm{e}^u\left(\frac{\pi}{2}+\frac{\pi}{2}\right)\mathrm{d}u=\int_{-1}^{0}\pi\mathrm{e}^u\mathrm{d}u=\pi\left(1-\frac{1}{\mathrm{e}}\right).$$

21.【解析】方法一 用极坐标系.

将$(x-1)^2+y^2\leqslant 1$，即$x^2+y^2\leqslant 2x$，转化成$r^2\leqslant 2r\cos\theta$，即$r\leqslant 2\cos\theta$.由此，$\cos\theta\geqslant 0$，从而$-\frac{\pi}{2}\leqslant\theta\leqslant\frac{\pi}{2}$，所求积分

$$I=\iint_D(x^2-3y^2)\mathrm{d}x\mathrm{d}y=\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\mathrm{d}\theta\int_0^{2\cos\theta}(r^2\cos^2\theta-3r^2\sin^2\theta)r\mathrm{d}r$$

$$=4\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}(\cos^2\theta-3\sin^2\theta)\cos^4\theta\mathrm{d}\theta=4\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\left[4\cos^2\theta-(3\cos^2\theta+3\sin^2\theta)\right]\cos^4\theta\mathrm{d}\theta$$

$$=4\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}(4\cos^6\theta-3\cos^4\theta)\mathrm{d}\theta=8\int_0^{\frac{\pi}{2}}(4\cos^6\theta-3\cos^4\theta)\mathrm{d}\theta$$

$$=8\left(4\times\frac{5}{6}\times\frac{3}{4}\times\frac{1}{2}\times\frac{\pi}{2}-3\times\frac{3}{4}\times\frac{1}{2}\times\frac{\pi}{2}\right)=\frac{\pi}{2}.$$

方法二 平移坐标系.

将坐标系整体向右平移一个单位，积分区域变成$D'=\{(x,y)\,|\,x^2+y^2\leqslant 1\}$，此时所求积分

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$$I=\iint_{D'}\left[(x+1)^2-3y^2\right]\mathrm{d}x\mathrm{d}y=\iint_{D'}(x^2+2x+1-3y^2)\mathrm{d}x\mathrm{d}y.$$

根据对称性，$\iint_{D'}2x\mathrm{d}x\mathrm{d}y=0$；根据轮换对称性，$\iint_{D'}x^2\mathrm{d}x\mathrm{d}y=\iint_{D'}y^2\mathrm{d}x\mathrm{d}y$.故

$$\iint_{D'}(x^2-3y^2)\mathrm{d}x\mathrm{d}y=-\iint_{D'}(x^2+y^2)\mathrm{d}x\mathrm{d}y=-\int_{-\pi}^{\pi}\mathrm{d}\theta\int_0^1 r^2\cdot r\mathrm{d}r=-2\pi\times\frac{1}{4}=-\frac{\pi}{2},$$

进而 $I=\pi-\frac{\pi}{2}=\frac{\pi}{2}$.

【注】方法一的最后一步可以通过反复进行分部积分得到，但华里士公式作为熟知公式，应当直接记住.

方法二之所以能够进行，一个重要的前提是被积函数在平移之后仍然简单.如果被积函数在平移变换之后复杂程度明显上升，那么转化积分区域可能是得不偿失的.

22.【解析】由 $x^2+y^2\leqslant x+y$，得 $\left(x-\frac{1}{2}\right)^2+\left(y-\frac{1}{2}\right)^2\leqslant\frac{1}{2}$.

令 $\begin{cases}x=\frac{1}{2}+r\cos\theta,\\ y=\frac{1}{2}+r\sin\theta,\end{cases}$ 其中 $0\leqslant\theta\leqslant2\pi$，$0\leqslant r\leqslant\frac{1}{\sqrt{2}}$，则

$$\iint_D(x+y^2)\mathrm{d}\sigma=\int_0^{2\pi}\mathrm{d}\theta\int_0^{\frac{1}{\sqrt{2}}}\left[\left(\frac{1}{2}+r\cos\theta\right)+\left(\frac{1}{2}+r\sin\theta\right)^2\right]r\mathrm{d}r$$

$$=\int_0^{2\pi}\mathrm{d}\theta\int_0^{\frac{1}{\sqrt{2}}}\left(\frac{3}{4}+r\cos\theta+r\sin\theta+r^2\sin^2\theta\right)r\mathrm{d}r$$

$$=\int_0^{2\pi}\left[\left(\frac{3}{8}r^2+\frac{1}{3}r^3\cos\theta+\frac{1}{3}r^3\sin\theta+\frac{1}{4}r^4\sin^2\theta\right)\Big|_{r=0}^{r=\frac{1}{\sqrt{2}}}\right]\mathrm{d}\theta$$

$$=\int_0^{2\pi}\left[\frac{3}{16}+\frac{1}{6\sqrt{2}}(\sin\theta+\cos\theta)+\frac{1}{16}\sin^2\theta\right]\mathrm{d}\theta$$

$$=\frac{3}{8}\pi+\frac{1}{6\sqrt{2}}(\sin\theta-\cos\theta)\Big|_0^{2\pi}+\frac{1}{16}\int_0^{2\pi}\frac{1-\cos2\theta}{2}\mathrm{d}\theta$$

$$=\frac{3}{8}\pi+\frac{\pi}{16}=\frac{7}{16}\pi.$$

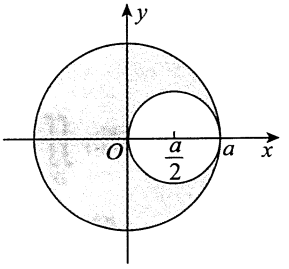
23.【解析】方法一　如图所示，积分区域 D 由两个圆周围成，并且被积函数是因子 x^2+y^2 的函数，故选用极坐标系计算.

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$$I=\iint_D r\cdot r\mathrm{d}r\mathrm{d}\theta=\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\mathrm{d}\theta\int_{a\cos\theta}^{a}r^2\mathrm{d}r+\int_{\frac{\pi}{2}}^{\frac{3\pi}{2}}\mathrm{d}\theta\int_0^a r^2\mathrm{d}r$$

$$=\frac{2}{3}\int_0^{\frac{\pi}{2}}(a^3-a^3\cos^3\theta)\mathrm{d}\theta+\frac{\pi}{3}a^3$$

$$=\frac{2\pi}{3}a^3-\frac{2}{3}a^3\int_0^{\frac{\pi}{2}}\cos^3\theta\mathrm{d}\theta=\frac{2}{3}a^3\left(\pi-\frac{2}{3}\right).$$



方法二 由积分区域的可加性计算.

$$I=\iint_{x^2+y^2\leqslant a^2}\sqrt{x^2+y^2}\mathrm{d}x\mathrm{d}y-\iint_{\left(x-\frac{a}{2}\right)^2+y^2\leqslant\frac{a^2}{4}}\sqrt{x^2+y^2}\mathrm{d}x\mathrm{d}y$$

$$=\int_0^{2\pi}\mathrm{d}\theta\int_0^a r^2\mathrm{d}r-\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\mathrm{d}\theta\int_0^{a\cos\theta}r^2\mathrm{d}r$$

$$=\frac{2}{3}a^3\left(\pi-\frac{2}{3}\right).$$

24. 【解析】令 $\begin{cases}x=r\cos\theta,\\ y=r\sin\theta,\end{cases}$ 其中 $\begin{cases}0\leqslant\theta\leqslant\dfrac{\pi}{2},\\ 0\leqslant r\leqslant 1,\end{cases}$ 则有

$$\iint_D\sqrt{\frac{1-x^2-y^2}{1+x^2+y^2}}\mathrm{d}x\mathrm{d}y=\int_0^{\frac{\pi}{2}}\mathrm{d}\theta\int_0^1\sqrt{\frac{1-r^2}{1+r^2}}r\mathrm{d}r$$

$$=\frac{\pi}{2}\cdot\frac{1}{2}\int_0^1\sqrt{\frac{1-r^2}{1+r^2}}\mathrm{d}(r^2)\overset{令r^2=t}{=\!=\!=}\frac{\pi}{4}\int_0^1\sqrt{\frac{1-t}{1+t}}\mathrm{d}t$$

$$=\frac{\pi}{4}\int_0^1\frac{1-t}{\sqrt{1-t^2}}\mathrm{d}t=\frac{\pi}{4}\left(\int_0^1\frac{\mathrm{d}t}{\sqrt{1-t^2}}-\int_0^1\frac{t}{\sqrt{1-t^2}}\mathrm{d}t\right)$$

$$=\frac{\pi}{4}(\arcsin t+\sqrt{1-t^2})\Big|_0^1=\frac{\pi}{8}(\pi-2).$$

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